	induced in the coil increases,	
7(a)(i)	r.m.s. current $I_{rms} = \dfrac{I_o}{\sqrt{2}} = \dfrac{2}{\sqrt{2}}$ $= 1.4 \text{ A}$	[1] substitution [1] answer
7(a)(ii)	r.m.s. voltage $V_{rms} = \dfrac{V_o}{\sqrt{2}}$ $= \dfrac{240}{\sqrt{2}} \text{ V}$ Mean power supplied $\langle P \rangle = I_{rms} V_{rms}$ $\langle P \rangle = \left(\dfrac{2}{\sqrt{2}}\right)\left(\dfrac{240}{\sqrt{2}}\right)$ $= 240 \text{ W}$ **OR** Peak power $P_o = V_o I_o$ = (240)(2) 480 W $\langle P \rangle = \dfrac{P_o}{2} = \dfrac{480}{2} = 240 \text{ W}$	[1] substitution [1] answer

No.	Solution	Remarks
7(b)(i)	$\frac{V_2}{V_1} = \frac{N_2}{N_1} \rightarrow \frac{V_2}{240} = \frac{10}{500} \rightarrow V_2 = 4.8 \text{ V}$ $\frac{V_1}{V_2} = \frac{I_2}{I_1} = \frac{N_1}{N_2} \rightarrow \frac{I_2}{2} = \frac{500}{10} \rightarrow I_2 = 100 \text{ A}$ Assumption: Transformer is 100% efficient / ideal with zero ohmic, magnetic leakage, copper and core losses. The output power is equal to the input power of the transformer.	[1] answer [1] answer [1] reasonable answer
7(b)(ii)	Welding Induction furnace Smelting scrap metal	[1] reasonable answer
8(a)(i)	When the vibrator oscillates the string, the waves travel along	[1] reflection at